Biomolecular Condensates: Nature's Smart Compartments

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Summary

- Equilibrium liquid-liquid phase separation principles apply within living cells.
- Slowly diffusing multivalent complexes drive local droplet condensation.
- Phase diagrams show spinodal decomposition in deeply phase-separated regions and nucleation near the binodal boundary.
- DNA and mRNA act as potent phase separation nucleators.
- The Corelet system enables studying phase separation and developing optogenetic tools for therapeutic use.
- Changes in surface tension cause immiscible multiphase condensates.

Introduction

Cells regulate their internal environment through specialized compartments, some of which lack membranes and form via protein phase separation [1]. This process drives the formation of biomolecular condensates essential for cellular functions. However, the precise localization mechanisms, particularly for low-abundance proteins, remain unclear [2]. Recent studies suggest that proteins and RNAs undergo liquid-liquid phase separation (LLPS) at specific intracellular sites by interacting with slow-diffusing nucleation centers [1]. These weak multivalent interactions, often involving intrinsically disordered regions (IDRs) and nucleic acids like RNA, facilitate condensate formation. IDRs, with their flexible conformations and multiple binding sites, serve as ideal scaffolds for this process.

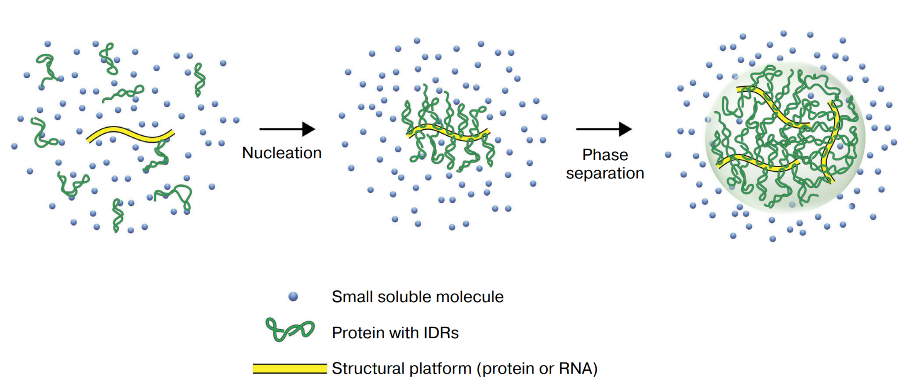

Figure 1. Proteins containing IDRs LLPS. Figure from [3].

LLPS creates unique material properties, particularly interfacial tension between dense and dilute phases, resulting in spherical droplets that merge and relax. Abnormal condensate behavior is linked to various diseases, making it crucial to understand their material properties, composition, and phase behavior. This knowledge is key to deciphering cellular organization and developing therapies for age-related diseases. However, reliable methods for predicting these properties and mapping LLPS in cells remain limited.

Experimental Setup

Engineered proteins consisting of a light-activatable 24-mer ferritin core (ferritin-iLID) and IDRs fused to SspB (SspB-FUS$_N$) enabled studies of LLPS [2]. Upon blue light activation, SspB and iLID bind, causing the core and IDRs to form clusters (Figure 2 A), mimicking LLPS in natural cells.

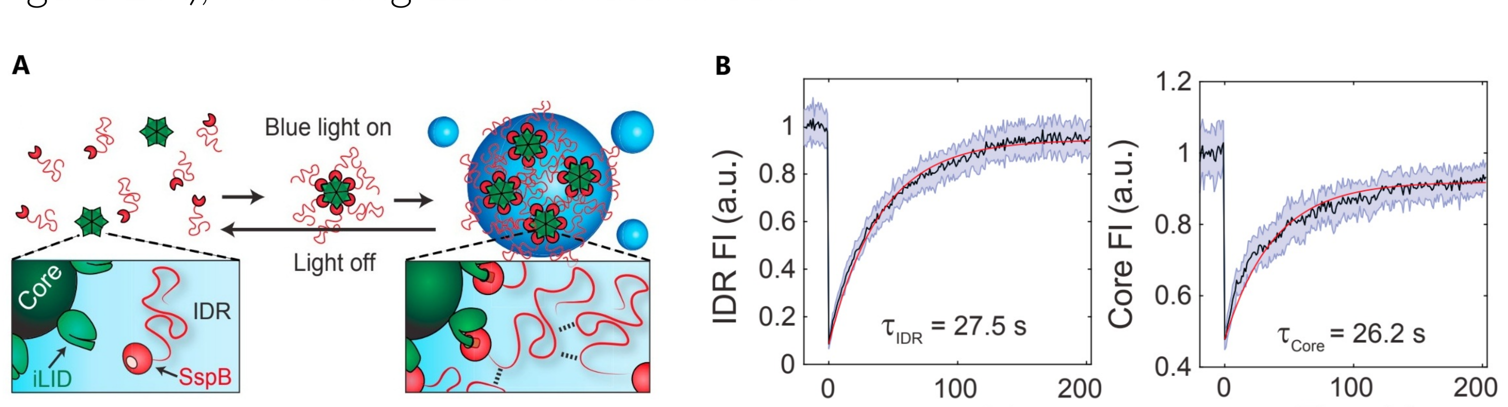

Figure 2. A: Photo-activated FUS$_N$ Corelet system. B: FRAP of core and IDR. Figure from [2].

Results and Discussion

To test LLPS through Corelet activation, we can use cells with varying expression levels of FUS$_N$-SspB and 24-mer NLS Ferritin-iLID as proposed in [2] (Figure 2), measuring their nuclear concentration ratio ($\overline{f}$). Before activation, Core concentration showed a unimodal distribution. After activation, phase-separating cells exhibited bimodal distributions, indicating the formation of two distinct phases (Figure 3 A). Cells with low $\overline{f}$ did not undergo phase separation, whereas those with higher $\overline{f}$ formed condensates with increasing sphericity suggesting that $\overline{f}$ represents an effective interaction strength between IDR-decorated cores.

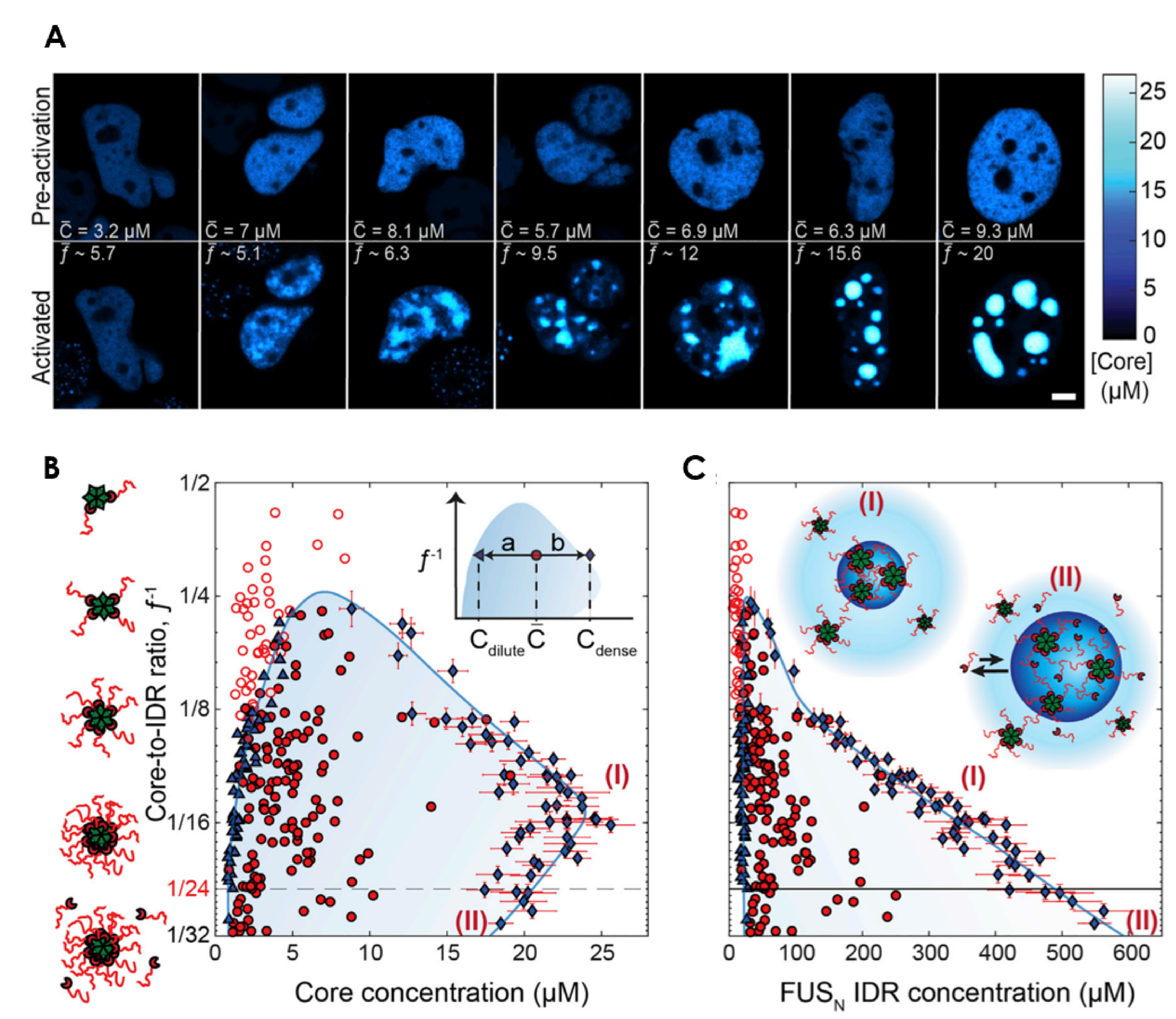

Figure 3. Phase diagram of FUS$_N$ Corelets and (A) confocal images of Cores in FUS$_N$ Corelets with respect to (B) Core concentration and Core-to-IDR ratio and (C) FUS$_N$ IDR concentration. Color bar shows FCS-based fluorescence-to-concentration conversion concentration. Data from [2].

Cells near the phase diagram peak form irregular condensates with undulating boundaries (Figure 3 B), indicating vanishing surface tension near a critical point. Core saturation shapes the phase diagram: as IDRs per core increase (lower $\overline{f}$), the binodal shifts to lower core concentrations (Figure 3 B). Plotting by IDR concentration eliminates this effect, showing a straight binodal edge, highlighting IDR-IDR interactions as key (Figure 3 C).

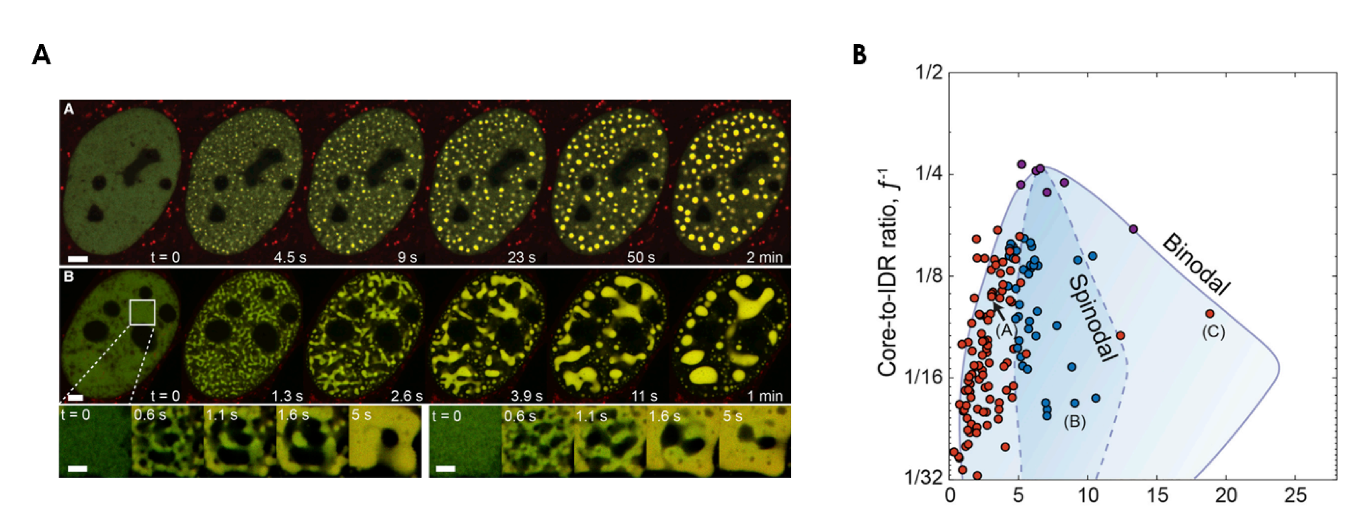

Figure 4. FUS$_N$ Corelets phase separation at (A) low and intermediate Core concentrations and (B) condensations modes mapped in the phase diagram. Data from [2].

Two-phase cells show network-like condensates, i.e. **spinodal decomposition**, while those near the binodal display punctate nucleation and growth (Figure 4 B).

Physical Principles of Biomolecular Condensates

Kinetic Framework of Light-Induced Phase Separation

Using the kinetic model presented in [1], the inactivated state transitions (C_i) to the activated state (C_a) follow a first-order reaction, with a rate proportional to blue light intensity: $k_1 = k_{act} \times [blue]$ where k_1 is an activation rate constant and $[blue]$ is the intensity of blue light. Activated molecules can also revert to the inactivated state at a rate k_2.

$$\frac{dC_i}{dt} = -k_1 C_i + k_2 C_a \tag{1}$$

$$\frac{dC_a}{dt} = k_1 C_i - k_2 C_a \tag{2}$$

Liquid-Liquid Phase Separation

To further explore the origins and dynamics of light-activated phase separation, we used a ternary regular solution model to describe the system's free energy [1] for the concentration of i, j populations ($\phi_{i,j}(\vec{r})$): activated, inactivated and other cytoplasmic molecules.

$$F[\phi_i(\vec{r})] = \int d\vec{r} \left[\sum_i \phi_i \ln \phi_i + \sum_{i \neq j} \chi_{ij} \phi_i \phi_j + \sum_i (\lambda_i \nabla \phi_i)^2 \right] \tag{3}$$

where $\chi_{ij}(\vec{r})$ define the strength of interaction between i and j molecules and λ_i is the surface energy coefficient for population i. Phase separation occurs when a sufficiently strong light stimulus drives an initially i-rich/j-poor system across the two-phase miscibility boundary leading to condensate growth.

A coarse-grained computer simulation modeled FIB1 and rRNA as linear polymers, and NPM1 as a branched polymer [4], resulting in a three-phase system (Figure 5 B). It was found that $\chi_{13} > \chi_{12} > \chi_{23}$ (see Eq. 3). This implies that $\gamma_{13} > \gamma_{12} > \gamma_{23}$, favouring FIB1 droplets embedded in NPM1 droplets, aligning with in vivo findings (Figure 5 A).

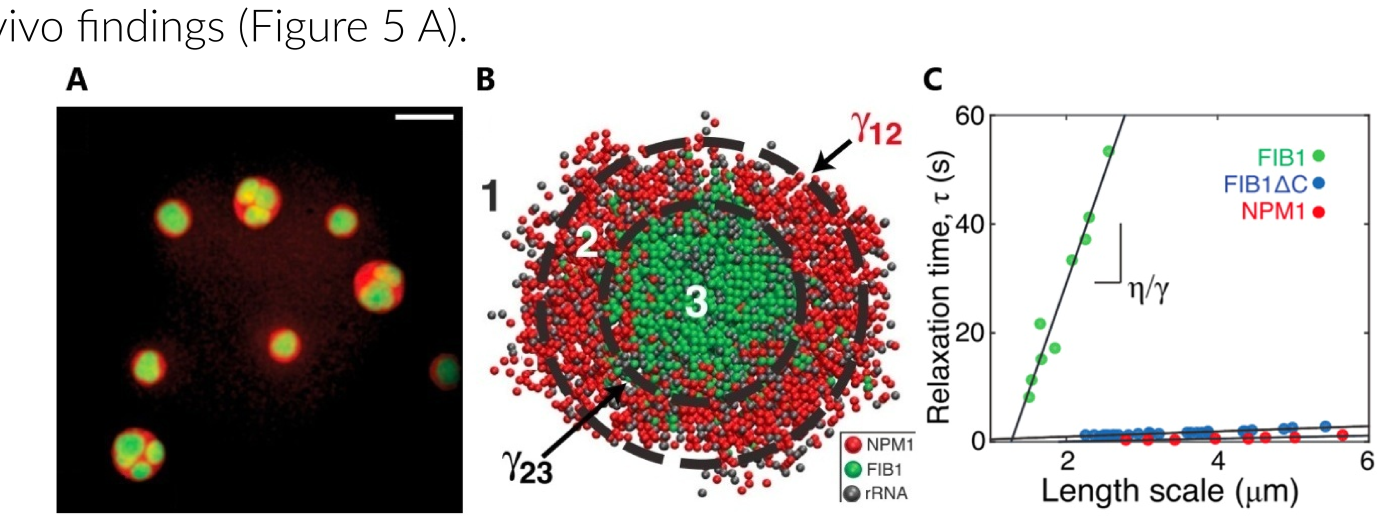

Figure 5. Immiscible multiphase droplets, A: in vivo. B: computer simulation. C: Relaxation time vs length scale for fusion of FIB1 and NPM1 droplets in vitro. Figure from [4].

The relaxation time τ for homotypic fusion varies with size and composition (Figure 5 C). NPM1 behaves as a liquid ($\eta = 0.74 \pm 0.06$ Pa·s), while FIB1 is viscoelastic.

References

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